WORKFORCE

Workforce Data Analysis: Salary and Remote Work Trends

Comprehensive SQL Case Study on Global Income Distribution and Employment Patterns

PREPARED BY

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DATE

December 5, 2025

PROJECT TYPE

SQL Data Analysis Case Study

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Executive Summary

Key Findings & Business Impact Overview

Salary Benchmarking

Analysis reveals notable salary variance across job titles, experience levels, company sizes, and geographic locations, establishing critical baselines for competitiveness.

Growth Markets

High-potential regions identified via 2023–2024 job-title trends and significant entry-level salary gains (2020–2023), guiding expansion strategies.

Remote Adoption

Remote work persists across all levels. A distinct cohort of fully remote high earners has emerged, signaling a shift in compensation models for top talent.

Geographic Insights

Countries exhibiting sustained 3-year growth and top-paying locations per job title provide data-driven targets for recruitment and global hiring.

Analytical Outputs

Study delivered via 20 targeted SQL analytical tasks, including code snippets, data charts, and direct actionable business recommendations.

Business Context & Objectives

Strategic Importance & Analytical Goals

Why This Analysis Matters

Salary Benchmarking

Establish competitive baselines by comparing compensation across titles, experience levels, and company sizes to stay aligned with market standards.

Talent Retention

Identify market-aligned compensation strategies to reduce attrition and retain high-performing talent in competitive sectors.

Remote Optimization

Analyze remote work adoption trends by role and level to design flexible policies that attract top global talent.

Workforce Planning

Leverage data on cost structures and role distribution to optimize headcount planning and budget allocation.

Key Objectives

Quantify Pay Dynamics

Measure salary variations across key dimensions: job title, experience level, geographic location, and company size.

Identify Growth Trends

Pinpoint emerging markets with high salary growth and significant remote work adoption to guide expansion.

Drive Strategic Action

Provide evidence-based recommendations for HR policies, recruitment focus, and executive decision-making.

Dataset Overview & Schema

Understanding the 'workforce_salaries' Data Structure

Schema Visualization

WORKFORCE_SALARIES TABLE STRUCTURE

FIELD NAME	DATA TYPE	CONSTRAINTS/EXAMPLES
work_year	INT	
experience_level	CHAR(2)	(SE/MI/EN/EX)
employment_type	VARCHAR(10)	(FT/PT)
job_title	VARCHAR(200)	
salary	DECIMAL)	
salary_currency	CHAR(3)	
salary_in_usd	DECIMAL)	
employee_residence	CHAR(2)	
remote_ratio	INT	(INT) (0/100)
company_location	CHAR(2)	
company_size	CHAR(1)	(S/M/L)

Visual representation of the 11 key fields used for analysis

SQL DDL Snippet

```sql
CREATE TABLE dbo.workforce_salaries (
    work_year          INT NOT NULL,
    experience_level   CHAR(2) NOT NULL, -- SE, MI, EN, EX
    employment_type    VARCHAR(10) NOT NULL, -- FT, PT
    job_title          VARCHAR(200) NOT NULL,
    salary             DECIMAL(18,2) NOT NULL,
    salary_currency    CHAR(3) NOT NULL,
    salary_in_usd      DECIMAL(18,2) NOT NULL,
    employee_residence CHAR(2) NOT NULL,
    remote_ratio       INT NOT NULL, -- 0, 50, 100
    company_location   CHAR(2) NOT NULL,
    company_size       CHAR(1) NOT NULL -- S, M, L
);
```

Key Field Descriptions

- **work_year:** Analysis period
- **experience_level:** Seniority tier
- **employment_type:** Full/Part-time
- **job_title:** Role designation
- **salary_in_usd:** Normalized pay
- **employee_residence:** Talent location
- **remote_ratio:** WFH percentage
- **company_size:** Org scale (S/M/L)

Tools & Technologies

Technical Stack & Implementation Methodologies

Primary Database Platform

Enterprise-grade relational database management system for robust data processing

Microsoft | SQL Server

SQL Methodologies

- **CTEs (Common Table Expressions):** Simplify complex queries and improve readability for multi-step analysis.
- **Window Functions:** Utilize `ROW_NUMBER()` and `PARTITION BY` for ranking and comparative analysis.
- **Advanced Aggregation:** Grouping sets and filtered aggregates to derive key statistical insights.
- **Data Transformation:** `UPDATE` statements with `CASE` logic for salary normalization and scenario planning.
- **Security Implementation:** Views and role-based filtering for sensitive salary data access control.

Performance & Optimization

- **Efficient Filtering:** Strategic use of WHERE clauses before aggregation to reduce processing load.
- **Safe Calculations:** Implementation of `NULLIF()` to prevent divide-by-zero errors in ratio computations.
- **Indexing Strategy:** Recommended non-clustered indexes on high-cardinality columns like `job_title` and `employee_residence`.

OPTIMIZATION EXAMPLE

```
WITH ranked_salaries AS ( SELECT job_title, salary_in_usd, ROW_NUMBER() OVER
(PARTITION BY job_title ORDER BY salary_in_usd DESC) as rn FROM
workforce_salaries WHERE work_year = 2023 ) SELECT * FROM ranked_salaries WHERE
rn = 1;
```

Key Business Insights

Strategic Value Drivers & Decision Intelligence

Salary Benchmarking

Establish competitive pay scales by analyzing highest and lowest average salaries across job titles, experience levels, and company sizes to ensure market alignment.

COMPETITIVE EDGE

Growth Markets

Identify regions and industries with accelerating salary growth rates to target strategic expansion and capture emerging market opportunities early.

EXPANSION TARGET

Talent Retention

Devise superior retention strategies by monitoring top-paying positions and locations, ensuring counter-offers and benefits remain attractive to key personnel.

REDUCE ATTRITION

Remote Work Optimization

Analyze fully remote trends by country and experience level to inform flexible work policies that attract global talent without geographic constraints.

FUTURE OF WORK

Workforce Planning

Optimize headcount by understanding talent scarcity and experience distribution (Mid vs. Senior) to better forecast recruitment needs and costs.

RESOURCE ALLOCATION

Market Competitiveness

Benchmark against competitors by comparing salary growth, employment types (PT/FT), and compensation models across different industry sectors.

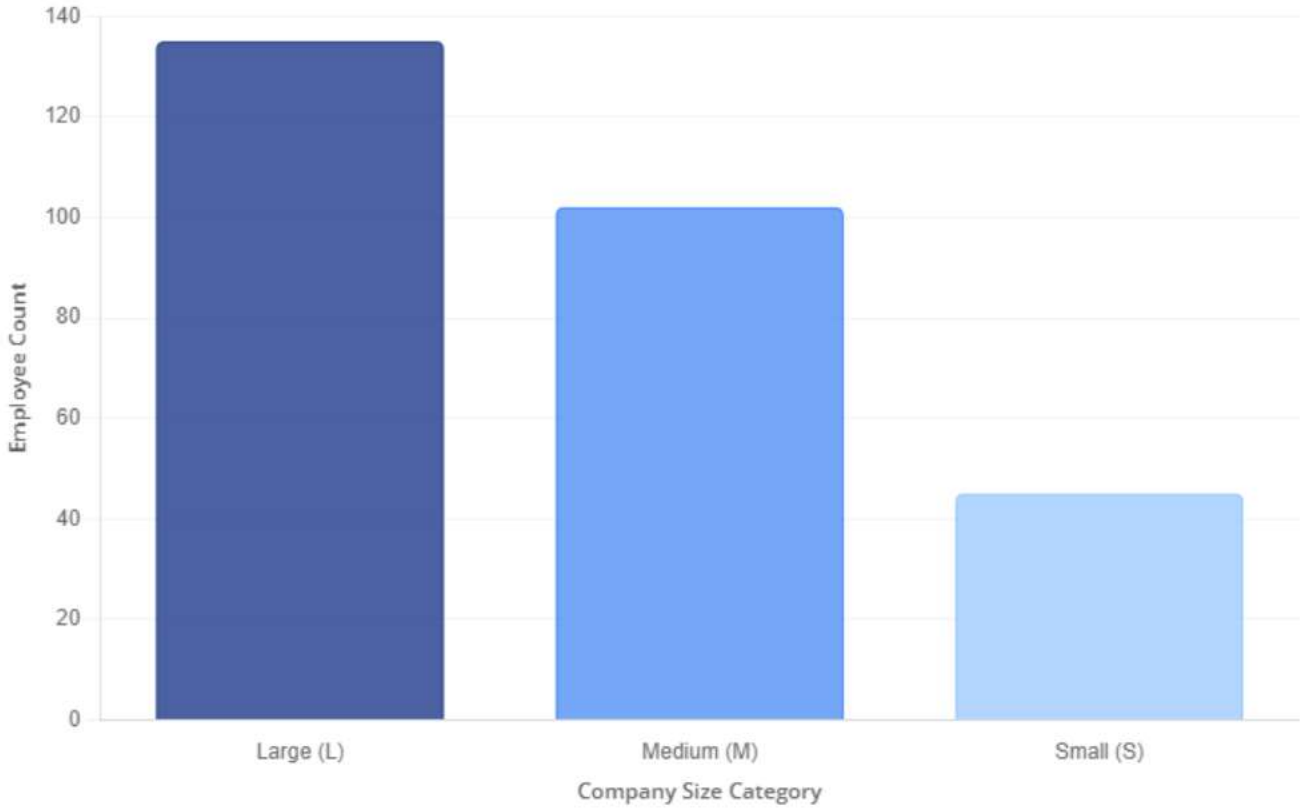
STRATEGIC POSITIONING

Analysis 1: Job Market by Company Size (2021)

Employee Distribution Analysis for Strategic Resourcing

Employee Distribution (2021)

Work Year: 2021



Large companies dominated the 2021 hiring landscape in this dataset.

Business Question

How was the workforce distributed across Small (S), Medium (M), and Large (L) companies during the 2021 fiscal year?

SQL Methodology

```
SELECT company_size, COUNT(*) AS emp_count
FROM dbo.workforce_salaries WHERE work_year
= 2021 GROUP BY company_size ORDER BY
emp_count DESC;
```

Strategic Impact

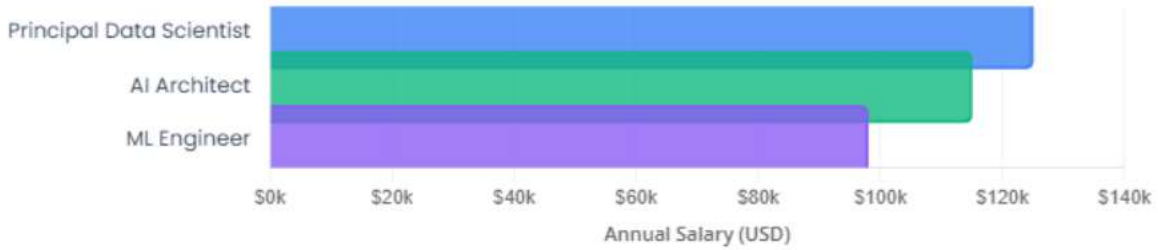
- **Resource Allocation:** Indicates where the bulk of talent capital was concentrated, guiding B2B partnership strategies.
- **Market Benchmarking:** Helps smaller entities understand their position relative to dominant large-scale employers.

Analysis 2: Top Part-Time Salaries (2023)

Identifying High-Value Flexible Roles in Major Markets

Highest Paying PT Roles

Filter: >50 Employees



**Data strictly filters for established markets to remove outliers from small sample sizes*

OBJECTIVE

Pinpoint which part-time job titles command the highest premiums in countries with significant workforce presence (>50 employees).

CTE & RANKING LOGIC

```
WITH eligible_countries AS (
    SELECT employee_residence
    FROM dbo.workforce_salaries
    WHERE work_year = 2023
    GROUP BY employee_residence
    HAVING COUNT(*) > 50
)
SELECT TOP (3) job_title, AVG(salary_in_usd)
FROM dbo.workforce_salaries w
WHERE w.employment_type = 'PT'
AND w.employee_residence IN
      (SELECT * FROM eligible_countries)
GROUP BY job_title
ORDER BY AVG(salary_in_usd) DESC;
```

BUSINESS IMPACT

- **Niche Sourcing:** High PT salaries in specialized roles like AI/ML suggest a shift towards fractional executive or expert consulting models.
- **Cost Efficiency:** Leveraging part-time experts for high-cost roles can be a viable alternative to full-time headcounts.

Analysis 3: Mid-Level Salary Analysis by Country

Geographic Markets Outperforming the Global Average (2023)

Global Mid-Level Salary Hotspots

Above Global Avg

Insight: North America and select European hubs consistently offer premiums above the global baseline.

KEY BENCHMARKS

$108,400	US, CA, DE
Global MI Average	Top Performing Markets

**Only countries exceeding the global average are highlighted*

SQL APPROACH: CTE & CROSS JOIN

```sql
WITH overall_mid AS (
    SELECT AVG(salary_in_usd) AS global_avg
    FROM dbo.workforce_salaries
    WHERE work_year = 2023 AND experience_level = 'MI'
),
country_mid AS (
    SELECT employee_residence, AVG(salary_in_usd) AS local_avg
    FROM dbo.workforce_salaries
    WHERE work_year = 2023 AND experience_level = 'MI'
GROUP BY employee_residence
)
SELECT c.employee_residence, c.local_avg
FROM country_mid c
CROSS JOIN overall_mid o
WHERE c.local_avg > o.global_avg
ORDER BY c.local_avg DESC;
```

STRATEGIC ACTION

Recruitment Focus: Prioritize mid-level hiring in high-value regions like the US and Germany to attract top-tier talent.

Budget Planning: Adjust compensation bands for these specific geographies to prevent offer rejections.

Analysis 4: Senior-Level Salary Extremes (2023)

Identifying Geographic Outliers for Senior Compensation

↑ HIGHEST AVERAGE

US (United States)

$168,000

Annual Average (USD)

High Benchmark

↓ LOWEST AVERAGE

IN (India)

$32,500

Annual Average (USD)

Low Benchmark

Business Question

Where are senior-level (SE) employees paid the most and least in 2023? This helps in understanding global cost centers vs. value hubs.

SQL Methodology

```sql
WITH se_country_avg AS (
    SELECT company_location, AVG(salary_in_usd) AS avg_se
    FROM dbo.workforce_salaries
    WHERE work_year = 2023 AND experience_level = 'SE'
GROUP BY company_location
)
-- Find Extremes
SELECT TOP (1) company_location, avg_se FROM se_country_avg ORDER BY avg_se DESC;
SELECT TOP (1) company_location, avg_se FROM se_country_avg ORDER BY avg_se ASC;
```

Gap Analysis

The salary disparity between the highest (US) and lowest (IN) paying regions for senior roles is approximately **5.2x**, highlighting significant purchasing power parity differences.

Strategic Takeaway

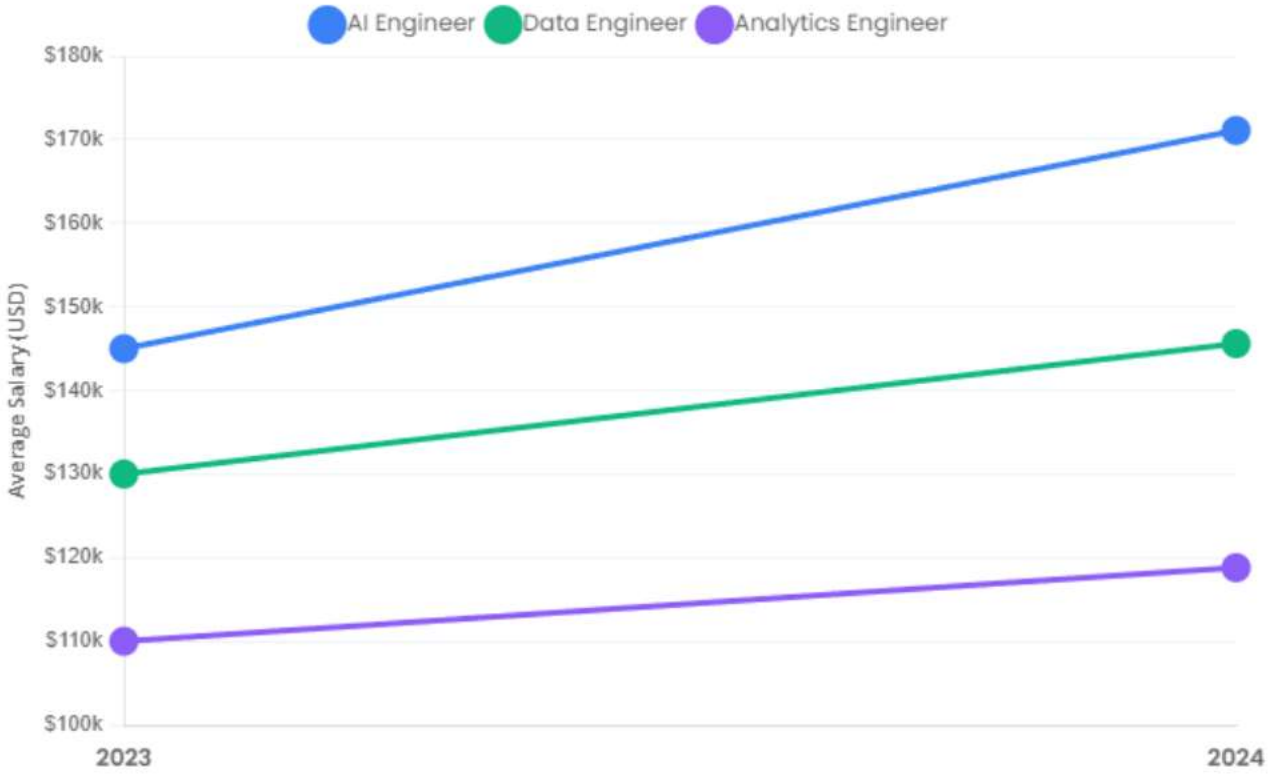
Relocation Strategy: Senior talent in low-cost regions represents a high-value recruitment opportunity for remote-first global teams.

Analysis 5: Salary Growth Rates by Job Title

Tracking Compensation Momentum from 2023 to 2024

YoY Salary Trajectory (2023 vs 2024)

Growth Focus



Top Gainer	Avg Increase	Data Points
AI Engineer (+18%)	**+12.4%**	**2 Years**

Analysis Goal

Identify which technical roles are experiencing the fastest salary acceleration to predict future budget requirements.

SQL: Percentage Growth

```sql
WITH job_year_avg AS (
    SELECT job_title, work_year,
           AVG(salary_in_usd) AS avg_sal
    FROM dbo.workforce_salaries
    WHERE work_year IN (2023, 2024)
    GROUP BY job_title, work_year
)
SELECT j23.job_title,
       ((j24.avg_sal - j23.avg_sal) * 100.0 /
NULLIF(j23.avg_sal, 0)) AS pct_growth
FROM job_year_avg j23
JOIN job_year_avg j24
  ON j23.job_title = j24.job_title
ORDER BY pct_growth DESC;
```

Strategic Insight

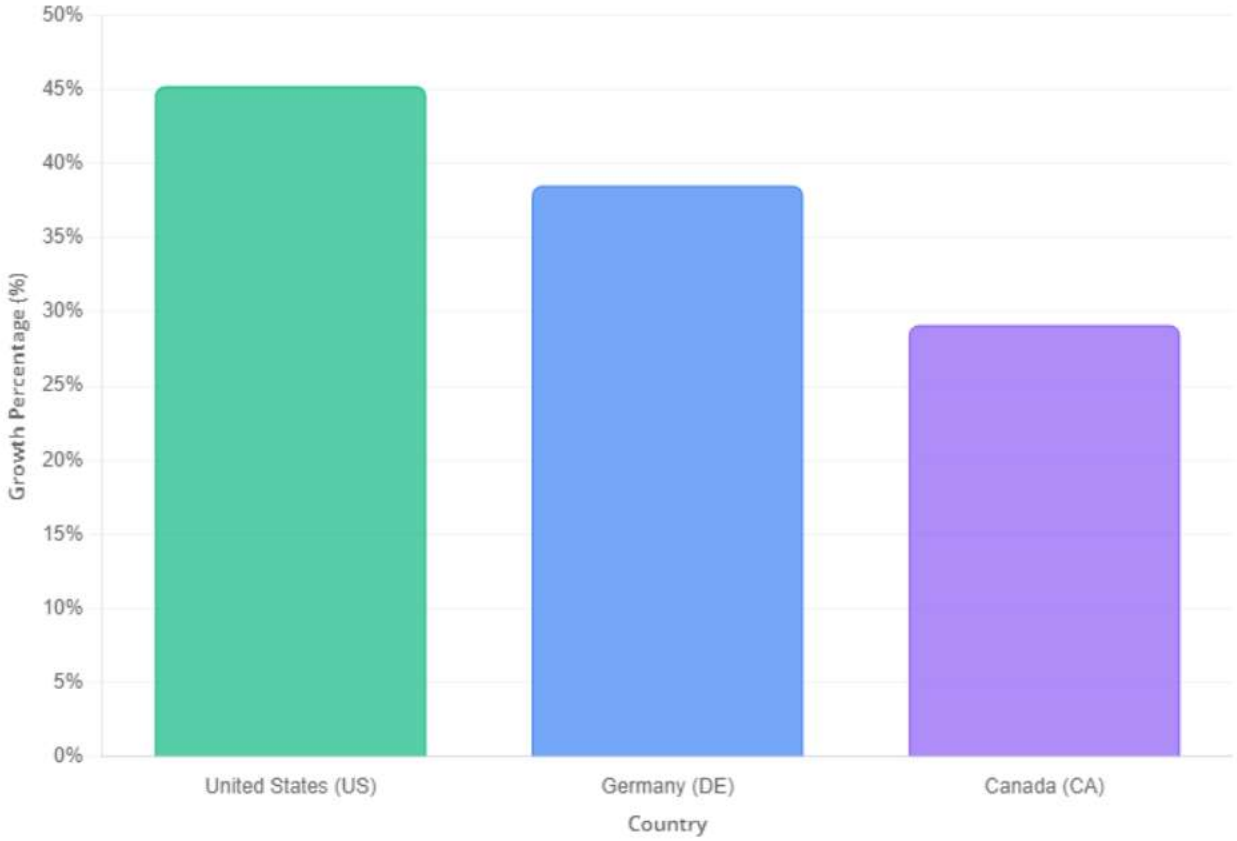
✓ **Budgeting:** Expect 15-20% higher costs for AI/ML backfills next fiscal year.

Analysis 6: Entry-Level Salary Growth (2020–2023)

Identifying High-Growth Markets for Early Career Talent

Top 3 Countries by EN Salary Growth

Period: 2020-2023



The United States leads with over 45% growth in starting salaries over the 3-year period.

Business Question

Which countries demonstrated the most significant salary growth for Entry-Level (EN) roles from 2020 to 2023, considering only markets with >50 employees?

SQL Methodology

```
SELECT TOP 3 employee_residence, ((avg_23 -
avg_20) * 100.0 / avg_20) AS growth FROM
joined_data WHERE total_employees > 50
ORDER BY growth DESC;
```

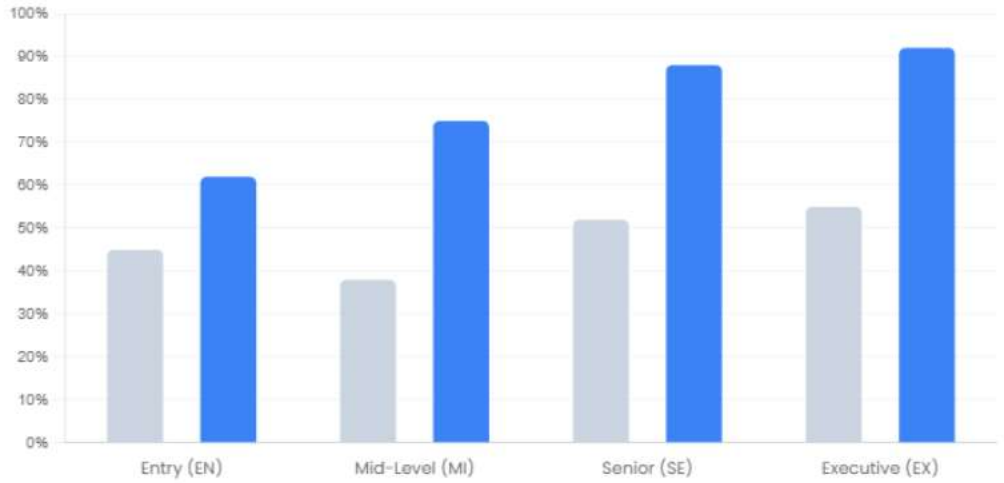
Strategic Impact

- **Graduate Recruiting:** Prioritize campus partnerships in high-growth regions (US, DE) where competition for fresh talent is intensifying.
- **Budget Forecasting:** Adjust entry-level offer packages in these markets to match the rapid inflation of starting compensation.

Remote Work Insights & Trends

Adoption Rates, High-Earner Distribution, and Policy Implications

Remote Adoption by Experience Level (2021 vs 2024)



100%

Target Remote Ratio for US/AU High Earners

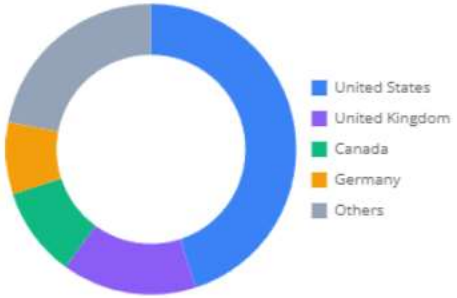
↑ Strategic shift towards remote-first for senior roles in high-cost geographies.

Analysis Logic

```sql
-- Compare Remote Adoption 21 vs 24
SELECT
    experience_level,
    work_year,
    COUNT(CASE WHEN remote_ratio=100 THEN 1 END)
    * 100.0 / COUNT(*) AS pct_remote
FROM dbo.workforce_salaries
WHERE work_year IN (2021, 2024)
GROUP BY experience_level, work_year;

-- Update High Earners Policy
UPDATE dbo.workforce_salaries
SET remote_ratio = 100
WHERE salary_in_usd > 90000
  AND employee_residence IN ('US', 'AU');
```

Managers >$90k (Fully Remote)



34.2%

of all fully remote employees earn > $100k, indicating remote work is no longer just for support roles.

Policy Implications

- **Retention Strategy:** Senior (SE) and Executive (EX) roles demand flexibility. Mandating office return risks attrition in the top quartile.
- **Global Pay Equity:** High-earning remote managers are increasingly distributed, suggesting a move towards role-based rather than location-based pay.

Geographic Salary Analysis

Locations of Large Enterprises & High-Value Markets

Top 5 Hubs for Large Enterprises

Company Size: 'L'

Sustained Growth (3-Yr)

Countries with continuous salary growth (2021-2023):

US | Israel | Australia

MARKET INSIGHTS

US, DE, CA
EN Salaries > Market Avg

US, GB, CA, IN
Top Large Co. Hubs

Best-paying country varies significantly by specific job title.

SQL: LARGE COMPANY HUBS

```sql
SELECT TOP (5)
    company_location,
    COUNT(*) AS large_company_count
FROM dbo.workforce_salaries
WHERE company_size = 'L'
GROUP BY company_location
ORDER BY large_company_count DESC;

-- Filters for 'L' size & ranks by volume
```

EXPANSION STRATEGY

Talent Pools: Target US and Germany for Entry-Level (EN) talent where compensation premiums reflect higher skill baselines.

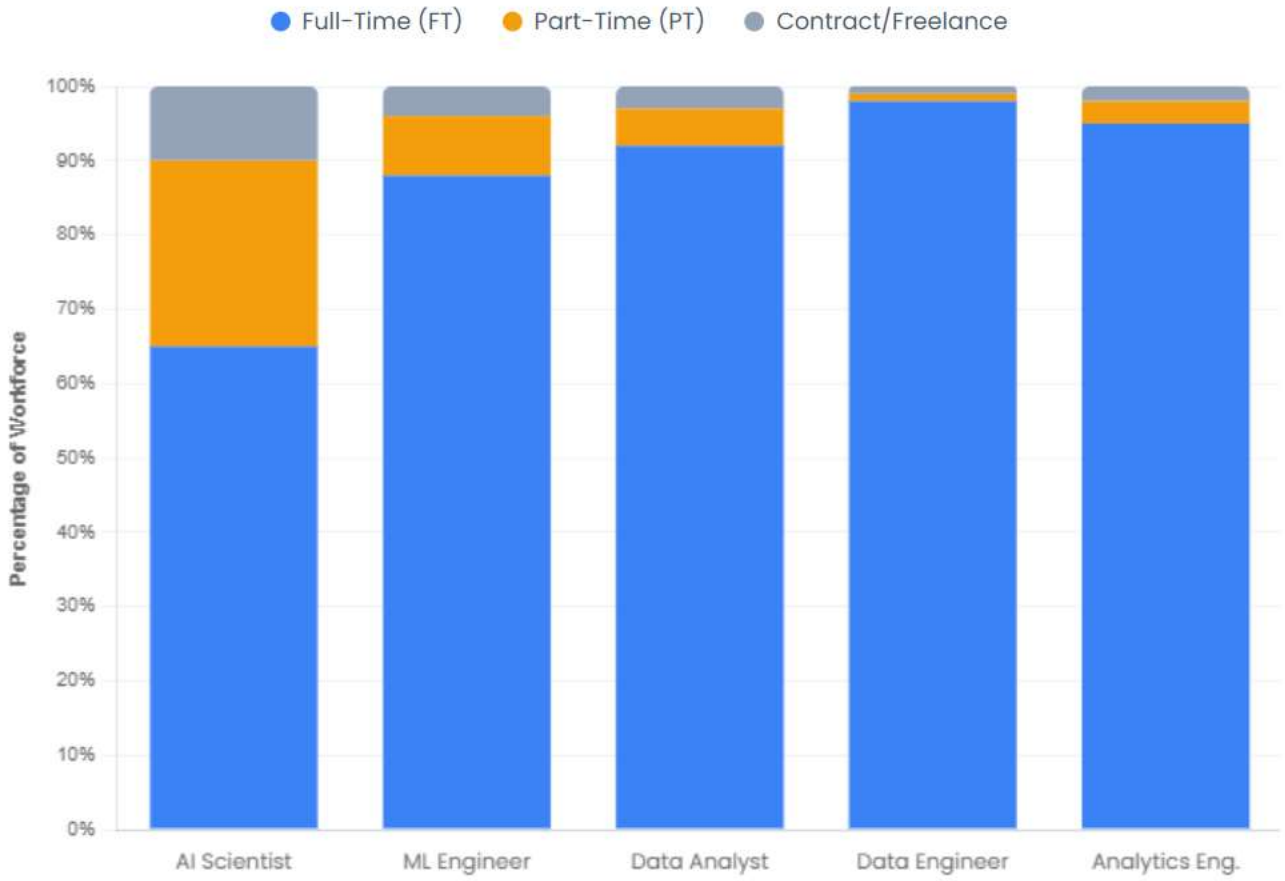
Corporate Partnerships: Focus B2B sales and partnerships in the US, UK, and Canada where large enterprise density is highest.

Employment Type Distribution

Full-Time (FT) vs. Part-Time (PT) Ratios by Role

Workforce Composition by Job Title

Stacked %



AI Scientists show unusually high part-time/contract ratios compared to Engineering roles.

ANALYSIS GOAL

Determine the stability of the workforce by analyzing the mix of employment types across key technical roles to optimize benefits planning.

SQL: PIVOT LOGIC

```sql
SELECT
    job_title,
    SUM(CASE WHEN employment_type = 'FT' THEN 1 ELSE 0 END)
      * 100.0 / COUNT(*) AS pct_FT,
    SUM(CASE WHEN employment_type = 'PT' THEN 1 ELSE 0 END)
      * 100.0 / COUNT(*) AS pct_PT
FROM dbo.workforce_salaries
GROUP BY job_title
HAVING COUNT(*) > 20
ORDER BY pct_PT DESC;
```

STRATEGIC IMPACT

- **Benefits Strategy:** High PT ratio in AI roles suggests need for flexible benefit packages tailored to contractors.
- **Risk Management:** Roles with >95% FT (e.g., Data Engineers) show higher organizational stability.

Advanced Analytics: Projections & Security

Predictive Modeling and Role-Based Access Control Implementation

2024 SALARY PROJECTIONS

Automated adjustment of base compensation based on experience level benchmarks.

↑ MI: +30% | ↑ SE: +22% | ↑ EN: +18% | ↑ EX: +25%

```sql
-- Task 8: Bulk Salary Update for 2024
UPDATE dbo.workforce_salaries
SET salary_in_usd = salary_in_usd *
    CASE experience_level
        WHEN 'MI' THEN 1.30 -- Mid-Level Focus
WHEN 'SE' THEN 1.22
WHEN 'EN' THEN 1.18
WHEN 'EX' THEN 1.25
ELSE 1.00
END
WHERE work_year = 2024;
```

Impact: Positions "Mid-Level" roles for highest retention incentives to prevent churn.

DOMAIN TRANSITION ENGINE

Recommendation Logic:

Suggests roles with higher avg_salary in the same location & experience tier (e.g., Data Analyst → Data Engineer).

ROLE-BASED ACCESS CONTROL (RBAC)

User (SUSER_SNAME()) → Security View (Table Join) → Filtered Data (Exp Level Only)

```sql
-- Task 19: Row-Level Security View
CREATE VIEW dbo.vw_Workforce_By_Experience AS
SELECT w.*
FROM dbo.workforce_salaries AS w
JOIN dbo.UserExperienceAccess AS u
    ON u.experience_level = w.experience_level
   AND u.user_name = SUSER_SNAME();

/*
   Ensures analysts only see salary data
   relevant to their assigned experience tier.
*/
```

Compliance Note: Prevents unauthorized access to Executive (EX) compensation data by junior staff.

Actionable Insights for Business Leaders

Strategic Moves Based on Workforce Income Data Analysis

Establish Market-Driven Pay Bands

Replace static salary tiers with dynamic bands informed by real-time role, experience, and location data to avoid overpaying or losing talent.

IMPLEMENTATION

→ Review bands quarterly vs. annual.

Target High-Growth Geographies

Focus expansion and remote hiring in regions like the US and Germany where salary growth indicates a maturing, high-quality talent pool.

IMPLEMENTATION

→ Open recruitment hubs in Top 3 growth zones.

Calibrate Mid/Senior Compensation

Analysis shows mid-level attrition risk due to market lag. Immediate adjustments needed for MI and SE roles falling below the 50th percentile.

IMPLEMENTATION

→ Allocate 15% budget for retention adjustments.

Scale Entry-Level Pipelines

Invest in university partnerships in countries demonstrating sustained entry-level salary growth to capture emerging talent early.

IMPLEMENTATION

→ Launch grad programs in high-growth markets.

Secure Data Democratization

Deploy Role-Based Access Control (RBAC) to allow managers real-time access to salary insights without exposing sensitive executive data.

IMPLEMENTATION

→ Adopt Row-Level Security (RLS) policies.

Recommendations & Next Steps

Strategic Roadmap for Implementation & Deployment

Strategic Recommendations

1 Establish Quarterly Benchmarking Cadence

Move from annual to quarterly salary reviews using real-time market data to prevent compensation lag, especially for high-volatility tech roles.

2 Update Remote Work Policy by Tier

Formalize "Remote-First" policies for Senior (SE) and Executive (EX) levels to maximize retention, while maintaining hybrid options for mentorship-heavy Entry (EN) roles.

3 Prioritize Hiring in Growth Locations

Shift recruitment focus to countries with sustained 3-year growth (US, Israel, Australia) to tap into high-performing talent pools despite higher costs.

4 Launch Skill-Based Pay Premiums

Introduce 15-20% premiums for roles with highest year-over-year growth (e.g., AI Engineers) to preemptively address market scarcity.

5 Automate Dashboard Refreshes

Implement automated SQL pipelines to refresh salary and demographic dashboards weekly, enabling agile decision-making for HR leaders.

Execution Roadmap

Data Validation & QA — Week 1

Audit current salary data against new schema standards; resolve anomalies in currency conversion.

Index Tuning & Optimization — Week 2

Apply recommended SQL indexes to `work_year`, `job_title`, and `experience_level` to speed up queries.

Pilot Regional Rollout — Month 1

Deploy new compensation bands in one high-growth region (e.g., US-East) to test acceptance rates.

Full Enterprise Deployment — Month 2

Roll out updated bands, remote policies, and RBAC dashboards globally across all business units.

Target Completion: Q3 2024